Combination circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Reduce a circuit by identifying genuine series and parallel groups.

SUCCESS CRITERIA - I can...

1. Identify nodes before grouping resistors.
2. Reduce one group at a time.
3. Check current and voltage conservation in the original circuit.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “series” and “parallel circuit” to explain your answer. Sentence starters are optional.

series: connected along one unbranched current path

parallel circuit: branches connected across the same two nodes

node: a shared electrical connection point

01 Warm-up: bring back an earlier idea

Measured (I, V) pairs are (**0.10**, **3**), (**0.20**, **6**), (**0.30**, **9**) in A, V. **Find resistance** and **explain how the measurements support** treating the resistance as constant.

Word help: resistance = voltage divided by current for a stated operating condition

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

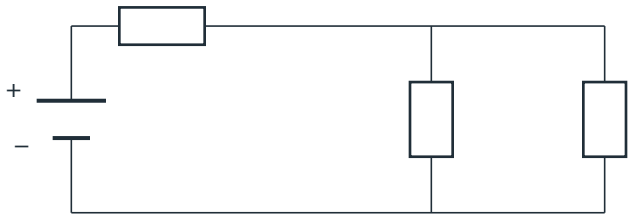
02 Worked example: explain the prediction

Reference: $R_{series} = R_1 + R_2$; $\mathbf{1}/R_{parallel} = \mathbf{1}/R_1 + \mathbf{1}/R_2$; $V = IR$. Reduce one known group at a time. **Check nodes before calling components** series or parallel.

Worked reasoning: A **2 ohm** resistor is in series with two **4 ohm** resistors in parallel. The parallel pair equals **2 ohms**; total **4 ohms**. A **12 V** supply gives **3 A** total.

Your turn: Draw the network and **label the two series sections** before splitting the branch current.

Sentence starter: In my work, I used … to show … . I labeled it … .



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A **2 Ω** resistor is in series with two parallel **6 Ω** resistors. **Draw the arrangement** and circle the parallel group.

Sentence starter: In my work, I used … to show … . I labeled it … .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A **10 V** supply powers a **2 Ω** resistor in series with two parallel **6 Ω** resistors.
1. **Label the network**.
2. **Find parallel equivalent resistance**, total current, series voltage drop, branch voltage and each branch current.

Sentence starter: I found … [unit]. This value means … in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A student adds all three resistor values because they are drawn side by side. What should find the grouping instead?

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Two targets each have range **600 m** but different radial velocities. **Explain how they can have** equal echo delays yet different Doppler shifts.
Label the outgoing and returning paths.

Sentence starter: Both … . They differ in … because … .

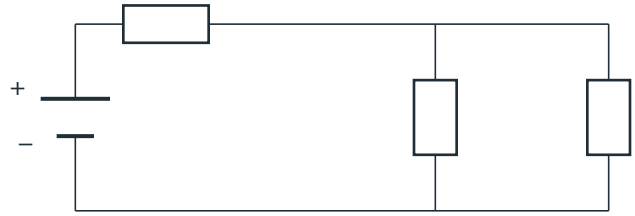


Marzano check - circle: ① ② ③

07 Exit check

Replace the two **6 Ω** branches with **3 Ω** and **6 Ω**.
1. Keep the **2 Ω** series resistor and **10 V** supply.
2. **Find total current**.
3. **Label the circuit** and **show the current path or** paths.

Sentence starter: I found … [unit]. This value means … in this situation.



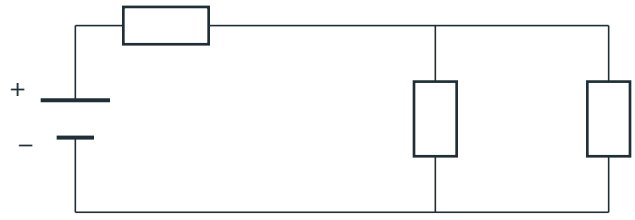
Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A **3 ohm** series resistor feeds parallel **6** and **3 ohm** branches from **15 V**.
1. **Find total current**, voltage across parallel section and each branch current.
2. **Label the circuit** and **show the current path or** paths.

Sentence starter: I found … [unit]. This value means … in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③